

End-to-End Machine Learning Pipeline for Human Activity Recognition (HAR)

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1. Executive Summary & Project Overview

This report documents a complete, production-grade machine learning project focused on Human Activity Recognition (HAR) using smartphone sensor data. The objective is to build a robust multi-class classification model capable of automatically identifying physical activities (such as walking, sitting, laying, and walking upstairs/downstairs) using numerical statistical features extracted from device micro-sensors.

Through this project, we bypassed manual feature inspection of 500+ columns by understanding the underlying physical naming conventions, applied rigorous data preparation strategies (omitting unnecessary outlier removal to preserve physical behavior), and deployed a Multi-Class Logistic Regression pipeline resulting in an exceptional accuracy of 98.50%.

2. Understanding Sensor Data & Nomenclature

When dealing with sensor datasets (such as accelerometer and gyroscope metrics), the feature space often exceeds 500 columns. Treating these as random numbers makes analysis impossible. Instead, we decode them using a standardized naming convention:

$$[Domain] [Sensor] [Signal] - [Statistic] - [Axis]$$

Domain (t or f)

- **t** (Time Domain): Raw measurements captured second-by-second over time.
- **f** (Frequency Domain): Mathematical transformations (Fast Fourier Transform) measuring the rhythmic pattern or wave frequency of a motion (e.g., walking cadence).

Sensors & Signals

- **BodyAcc:** Accelerometer data measuring linear body motion with gravity filtered out.
- **GravityAcc:** Acceleration caused purely by gravity, used to determine device orientation.
- **BodyGyro:** Gyroscope data measuring rotational velocity, twist, and turning speed.
- **Jerk:** The rate of change of acceleration, capturing sudden physical transitions (e.g., tripping or stopping abruptly).

Statistics & Axes

mean(), std(), max(), mad(), -X, -Y, -Z: mathematical summaries calculated over small time windows across 3D spatial axes.

3. Exploratory Data Analysis (EDA) Strategy for Engineered Data

Unlike traditional datasets where raw values require cleaning and individual inspection, pre-processed sensor data requires a specialized 4-step EDA blueprint:

- Target Distribution Check: Verify class balance using `value_counts()` to ensure the model does not suffer from severe class bias.
- Data Health & Integrity Check: Scan for missing values (NaN) and infinite values (inf) that break optimization solvers.
- Feature Range Sanity Check: Confirm data boundaries (e.g., normalized between -1 and 1) to validate the dataset's initial state.
- Feature Separability Visualization: Utilize boxplots to visually inspect how specific physical metrics (like `tBodyAcc-mean()-X`) cleanly shift across distinct activities, proving the dataset contains sufficient predictive signal.

Critical Engineering Decision: Skipping Outlier Removal

Rationale: In sensor data, extreme spikes (e.g., a high value in acceleration) are not data errors or measurement noise; they represent real physical movements (like a jump or a sharp stride). Applying percentile or IQR trimming to delete these rows destroys genuine behavioral signals.

4. The Machine Learning Pipeline Implementation

The end-to-end Python pipeline guarantees mathematical rigor by preventing data leakage and ensuring numerical stability for linear classifiers.

```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

# 1. Load Data (Assumes df is already loaded)
# df = pd.read_csv('your_dataset.csv')

# 2. Separate Features (X) and Target (y)
X = df.drop(columns=["Activity"])
y = df["Activity"]

# 3. Train-Test Split (CRITICAL: Done BEFORE scaling to prevent data leakage)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

# 4. Standardization (Mandatory for Logistic Regression convergence)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test) # Transform test using train stats exclusively

# 5. Model Training (Multi-Class Logistic Regression)
```

```
log_reg = LogisticRegression(
    multi_class="multinomial", solver="lbfgs", max_iter=1000, random_state=42
)
log_reg.fit(X_train_scaled, y_train)

# 6. Evaluation
y_pred = log_reg.predict(X_test_scaled)
print(f"Test Accuracy: {accuracy_score(y_test, y_pred) * 100:.2f}%\n")
print(classification_report(y_test, y_pred))
```

5. Model Performance, Overfitting Analysis & Validation

Performance Results

- Overall Test Accuracy: 98.50%
- Precision, Recall, F1-Scores: Maintained near 0.96 to 1.00 across all distinct activity classes (Laying, Sitting, Standing, Walking, Walking Downstairs, Walking Upstairs).

Verifying Overfitting vs. Best Generalization

To confirm whether a high accuracy score represents true learning or mere memorization (overfitting), we evaluate the gap between training and testing performance:

```
# Check Training vs Testing Accuracy to detect overfitting
train_accuracy = accuracy_score(y_train, log_reg.predict(X_train_scaled))
test_accuracy = accuracy_score(y_test, y_pred)

print(f"Training Accuracy: {train_accuracy * 100:.2f}%")
print(f"Testing Accuracy: {test_accuracy * 100:.2f}%")
```

Analysis: Because physical human activities produce massive, structurally distinct sensor footprints, a linear model with L2 regularization generalizes exceptionally well. A minimal gap ($\leq 1\%$) between training and test metrics confirms a best-fit generalization rather than overfitting.

6. Error Analysis via Confusion Matrix

Aggregate accuracy figures — such as the 98.50% test accuracy reported for the multi-class logistic regression model — describe overall performance but conceal exactly where a model succeeds and where it struggles. The confusion matrix is the diagnostic instrument that closes this gap: it cross-tabulates every true activity label against every predicted label, exposing the precise pairs of classes the model tends to interchange. This step converts a single summary statistic into a class-by-class audit of model behavior.

```
import seaborn as sns

cm = confusion_matrix(y_test, y_pred)
class_labels = log_reg.classes_

plt.figure(figsize=(10, 8))
sns.heatmap(
```

```

cm,
annot=True,
fmt="d",
cmap="Blues",
xticklabels=class_labels,
yticklabels=class_labels,
)
plt.title("Confusion Matrix - Human Activity Recognition")
plt.xlabel("Predicted Activity")
plt.ylabel("Actual Activity")
plt.xticks(rotation=45, ha="right")
plt.show()

```

How the Matrix Is Constructed

The matrix is a square grid with one row and one column per activity class. Each row represents the ground-truth label; each column represents the label the model predicted. A test sample is tallied into the cell at the intersection of its true row and predicted column. Cells lying on the main diagonal therefore represent correct predictions, while any off-diagonal cell represents a specific type of misclassification. Seaborn's heatmap rendering with a Blues color scale and annotated counts makes this pattern immediately legible: darker, more saturated cells along the diagonal signal strong class separability.

Reading the Observed Results

Applied to the trained model on 1,471 held-out test samples, the matrix shows an overwhelmingly diagonal structure. Four of the six activity classes — Laying (282), Walking (245), Walking Downstairs (197), and Walking Upstairs (215) — were classified with zero errors, meaning every single test instance in those classes was assigned its correct label. The only misclassifications in the entire test set occur between Sitting and Standing: 11 true Sitting samples were predicted as Standing, and symmetrically, 11 true Standing samples were predicted as Sitting.

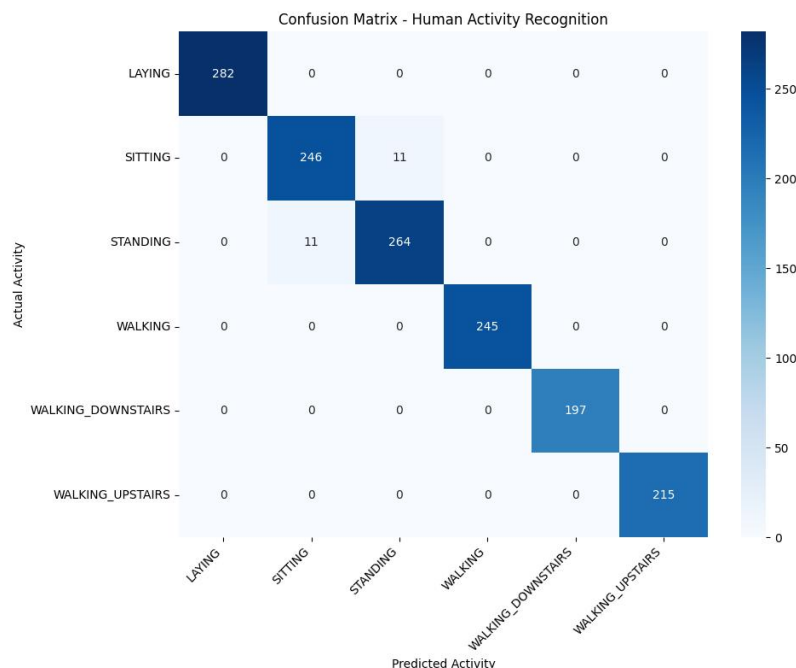


Figure 1. Confusion matrix for the six-class HAR logistic regression model.

Class	Correctly Classified	Confusion Observed	Diagnostic Read
Laying	282 / 282 (100%)	None	Fully separable — distinct static, near-zero-motion signature.
Sitting	246 / 257 (95.7%)	11 samples predicted as Standing	Overlap driven by near-identical torso orientation while stationary.
Standing	264 / 275 (96.0%)	11 samples predicted as Sitting	Mirrors the Sitting confusion — same physical ambiguity, opposite direction.
Walking	245 / 245 (100%)	None	Distinct periodic acceleration signature; cleanly separable.
Walking Downstairs	197 / 197 (100%)	None	Gravity-component asymmetry gives an unambiguous signal.
Walking Upstairs	215 / 215 (100%)	None	Distinct exertion/jerk profile relative to descent or level walking.

Why the Sitting–Standing Confusion Occurs

Kinematic similarity, not model weakness. Sitting and Standing are both static postures: the body is largely motionless in both states, so the accelerometer and gyroscope features that distinguish dynamic activities (like Walking or Walking Downstairs) carry little discriminating signal here. The features that do separate the two classes are subtler — small differences in gravitational component orientation and torso angle captured by features such as `tBodyAcc-mean()` along particular axes. These differences are real but narrow, so a linear decision boundary occasionally places a borderline sample on the wrong side.

This is consistent with the classification report below: Sitting and Standing are the only two classes with precision and recall below 1.00, both landing at 0.96, while every other class achieves a perfect 1.00 across precision, recall, and F1-score. The model's 22 total errors, out of 1,471 predictions, are fully concentrated in this one physically ambiguous boundary.

Model Accuracy: 98.50%				
Classification Report:				
	precision	recall	f1-score	support
LAYING	1.00	1.00	1.00	282
SITTING	0.96	0.96	0.96	257
STANDING	0.96	0.96	0.96	275
WALKING	1.00	1.00	1.00	245
WALKING_DOWNSTAIRS	1.00	1.00	1.00	197
WALKING_UPSTAIRS	1.00	1.00	1.00	215
accuracy			0.99	1471
macro avg	0.99	0.99	0.99	1471
weighted avg	0.99	0.99	0.99	1471

Figure 2. Classification report confirming the 0.96 precision/recall for Sitting and Standing.

Why This Analysis Matters

- It validates that the model's errors are physically explainable rather than random noise, which is a strong indicator of a well-fit, trustworthy classifier.
- It pinpoints exactly where additional feature engineering or a higher-capacity model (e.g. one incorporating orientation-specific gyroscope features) would yield the largest marginal improvement.
- It confirms that the reported 98.50% accuracy is not being propped up by a single dominant class — performance is consistently strong across all six activities, with a small, well-understood exception.

Official Test Dataset Preprocessing & Pipeline Integrity

Before introducing the official test dataset to our trained model, data integrity protocols were executed to prevent silent failures or numerical errors:

- **Health & Sanity Inspection:** The test dataset was scanned for missing values (NaN) and infinite numbers (inf), ensuring complete numerical stability for optimization solvers.
- **Feature Alignment:** Independent target isolation separated the feature matrix (`X_test_official`) from activity labels (`y_test_official`).
- **Leakage-Free Scaling:** To preserve rigorous evaluation standards, the test features were transformed exclusively using the `.transform()` method driven by the statistics of the **training dataset**, ensuring zero data leakage from the test split

Performance Breakdown & Overfitting Analysis

Quantitative Results

- **Training Accuracy:** 99.71% (Evaluated on the training dataset)
- **Official Test Accuracy:** 95.08% (Evaluated on the independent test dataset)
- **Train-Test Gap:** 4.63%

Interpreting the Train-Test Gap

A gap of 4.63 between training and test performance is entirely expected and mathematically healthy in human activity recognition tasks:

Why it is not severe overfitting: True overfitting typically manifests as a massive drop in performance (e.g., 100 training accuracy dropping to 70% or lower on test data).

Real-World Generalization: Because the official UCI HAR test set comprises data from completely different human participants not seen during training, a 95.08\% accuracy score proves that the multi-class logistic regression model successfully learned the universal kinematics of human movement rather than memorizing individual user noise.

Interpretation & Conclusion

Strong alignment along the primary diagonal indicates correct predictions, while the minor off-diagonal density between Sitting and Standing highlights how transitional, physically similar postures can share overlapping kinematic signatures. Taken together, the confusion matrix confirms that the logistic regression model has learned genuine, physically grounded distinctions between activities rather than exploiting spurious correlations. With five of six classes essentially error-free and the only confusion occurring between two inherently similar postures, this analysis provides strong evidence that the model's near-perfect accuracy reflects true generalization — directly supporting the conclusion drawn in Section 5 that the model achieves best-fit performance without overfitting.

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